Various derivatives of the function $f(x)$ evaluated at $x=2$ are given below.

Functions and its derivatives	$\frac{1}{x}$ **and its derivatives**	**value at** $x=2$
$f(x)$	$\frac{1}{x}$	$\frac{1}{2}$
$f'(x)$	$-\frac{1}{x^2}$	$-\frac{1}{4}$
$f''(x)$	$\frac{2}{x^3}$	$\frac{1}{4}$
$f'''(x)$	$-\frac{6}{x^4}$	$-\frac{3}{8}$

Table 7.4

Substituting these values, we get the required expansion of the function as

$$\frac{1}{x} = \frac{1}{2} - \frac{1}{4}\frac{(x-2)}{1!} + \frac{1}{4}\frac{(x-2)^2}{2!} - \frac{3}{8}\frac{(x-2)^3}{3!} + \cdots$$

which is, $$\frac{1}{x} = \frac{1}{2} - \frac{(x-2)}{4} + \frac{(x-2)^2}{8} - \frac{(x-2)^3}{16} + \cdots$$

EXERCISE 7.4

1. Write the Maclaurin series expansion of the following functions:

   (i) e^x (ii) $\sin x$ (iii) $\cos x$

   (iv) $\log(1-x)\,; -1 \le x < 1$ (v) $\tan^{-1}(x)\,; -1 \le x \le 1$ (vi) $\cos^2 x$

2. Write down the Taylor series expansion, of the function $\log x$ about $x=1$ upto three non-zero terms for $x > 0$.

3. Expand $\sin x$ in ascending powers $x - \frac{\pi}{4}$ upto three non-zero terms.

4. Expand the polynomial $f(x) = x^2 - 3x + 2$ in powers of $x - 1$.

7.5 Indeterminate Forms

In this section, we shall discuss various "indeterminate forms" and methods of evaluating the limits when we come across them.

7.5.1 A Limit Process

While computing the limits

$$\lim_{x \to \alpha} R(x)$$

of certain functions $R(x)$, we may come across the following situations like,

$$\frac{0}{0},\ \frac{\infty}{\infty},\ 0 \times \infty,\ \infty - \infty,\ 1^{\infty},\ 0^0,\ \infty^0.$$

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